Toward Accelerated LLM Inference: Porting and Evaluating Diffusion-Based Speculative Decoding on TPU

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Project Website

Problem & Motivation

- LLM decoding is **inherently sequential**: generating n tokens requires n target-model forward passes—latency grows linearly.
- **Speculative decoding** uses a lightweight *draft model* to propose candidates; the target verifies in one batched pass. Good drafts ⇒ multiple tokens accepted per step.
- **Our contribution:** Port DFlash—a diffusion-based drafter—from GPU/PyTorch to TPU/JAX inside `tpu-inference`. Benchmark Qwen3-4B across 9 datasets.
- **Key TPU advantage:** Verification cost is **flat from K=16 to K=128** (only 0.97×), while GPU scales to 2.3×.

Background

Speculative Decoding

- Draft n candidate tokens with a fast model, then verify all at once via rejection sampling.
- Key metric: τ = average accepted tokens per draft step.

Draft Model Approaches

- **N-gram:** Fast but low τ (poor distribution approximation).
- **Eagle3:** Autoregressive drafter, $O(k)$ sequential forward passes.
- **DFlash:** Diffusion-based *block drafter*—generates all 16 tokens in **one forward pass** via non-causal attention. Draft cost is $O(1)$ regardless of block size.

DFlash Architecture

- 4-layer transformer with **non-causal block attention**.
- Reuses target model's embedding layer and LM head.
- Conditioned on hidden states from layers [1, 9, 17, 25, 33], projected via FC + RMSNorm.
- Trained on only 289K samples; outperforms Eagle3 (1.4M samples).

Methods: GPU → TPU

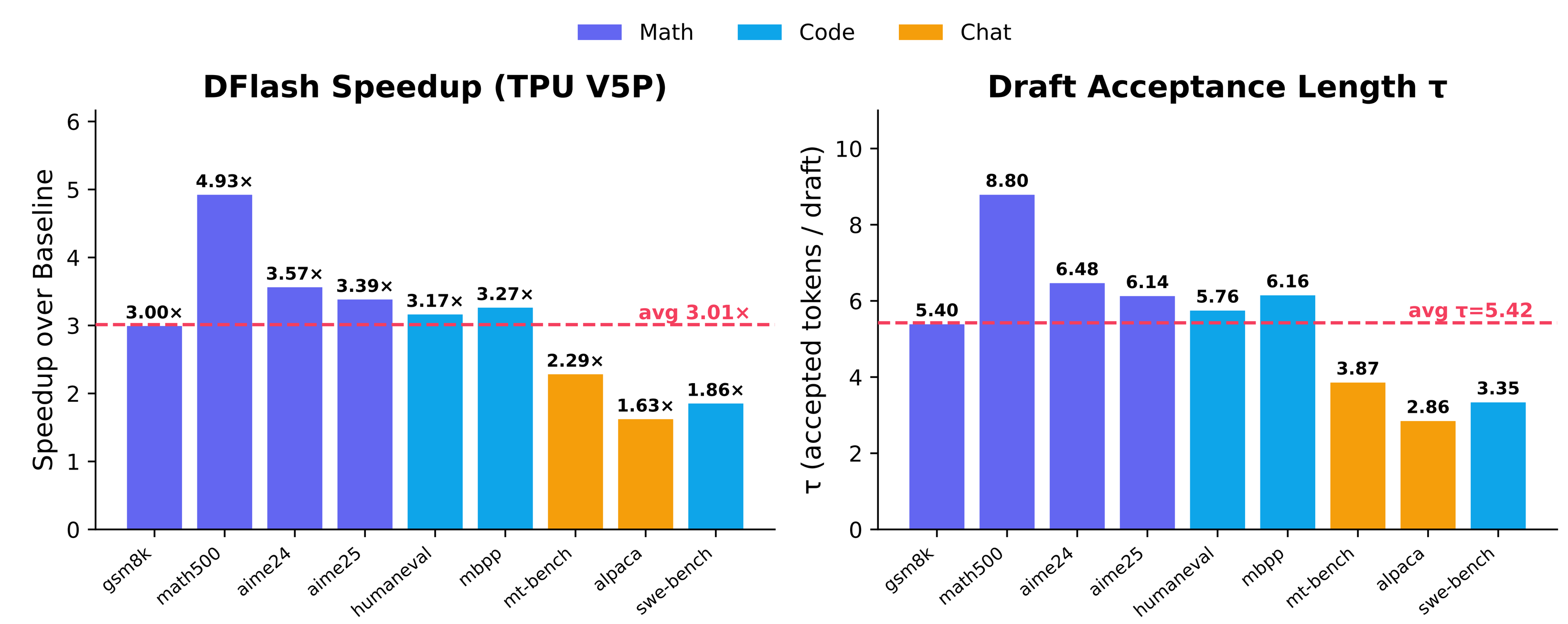
1. **Dual KV cache:** Paged KV (PagedAttention) for target; static JAX KV with `dynamic_update_slice` for draft.
2. **Non-causal attention:** Draft layers use TPU `flash_attention` with `causal=False`.
3. **Critical bug fix—sequence-length inflation:** Phantom tokens (∼15/step) corrupted RoPE embeddings. Fixing this *nearly doubled performance:*
   τ: 2.49 → 4.48 Speedup: 1.30× → 2.31×
4. **Hidden-state extraction:** Layers [1,9,17,25,33] concatenated, projected via FC + RMSNorm.
5. **Method registration:** New `"dflash"` branch in `tpu_runner.py` with prewarm support.

Zero new vLLM dependencies. Full DFlash contract preserved.

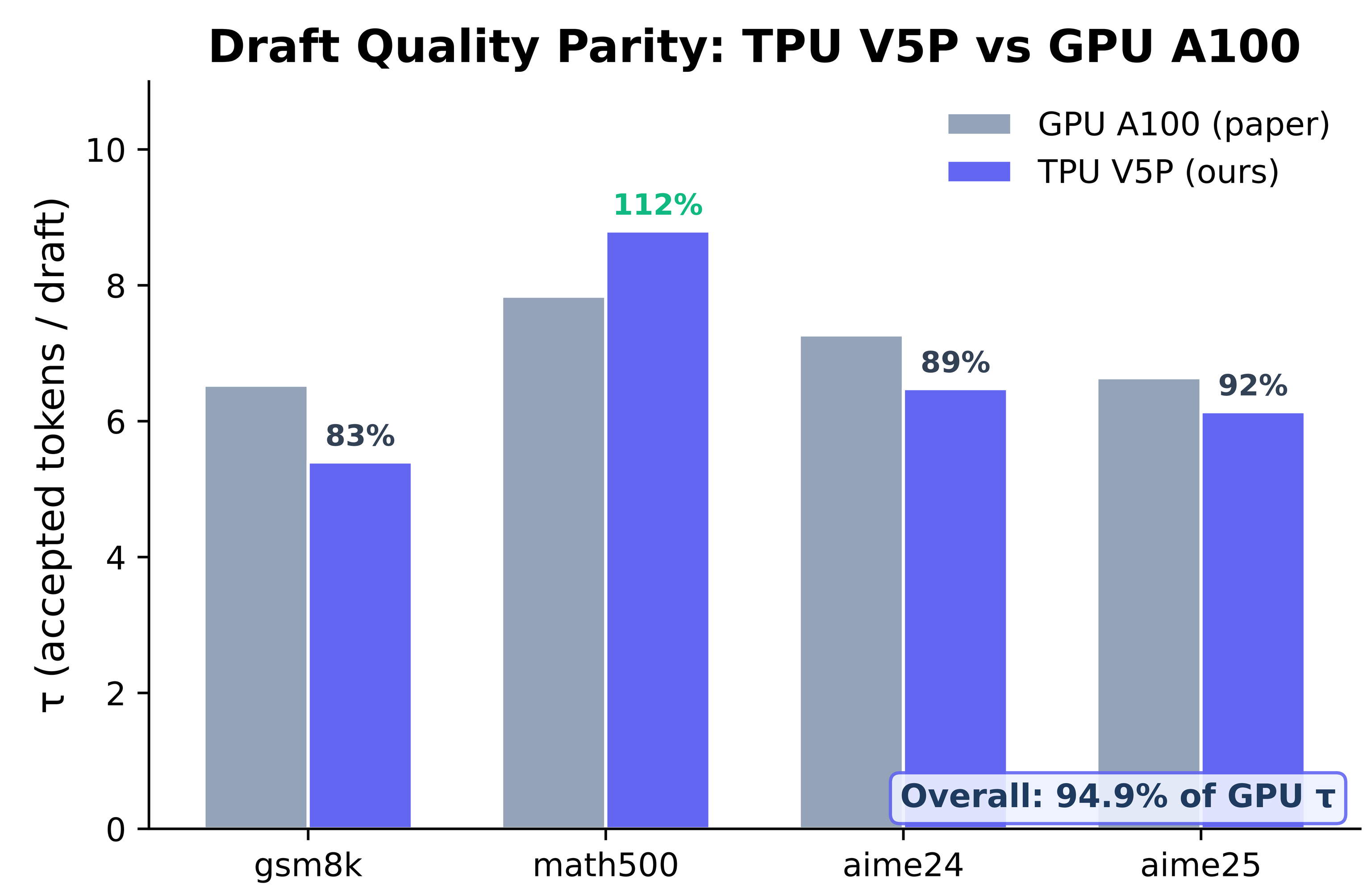
Results

2.31× vLLM pipeline **3.01×** Standalone V5P τ = **5.42** Avg. accepted

- Qwen3-4B + DFlash-b16 on **9 datasets**, TPU V5P (4 chips).
- Math: **3.72×** (τ=6.71) | Code: **2.77×** (τ=5.09) | Chat: **1.96×** (τ=3.37).
- Peak: Math500 at **4.93×** (τ=8.80), *exceeding* the GPU paper result (τ=7.84).

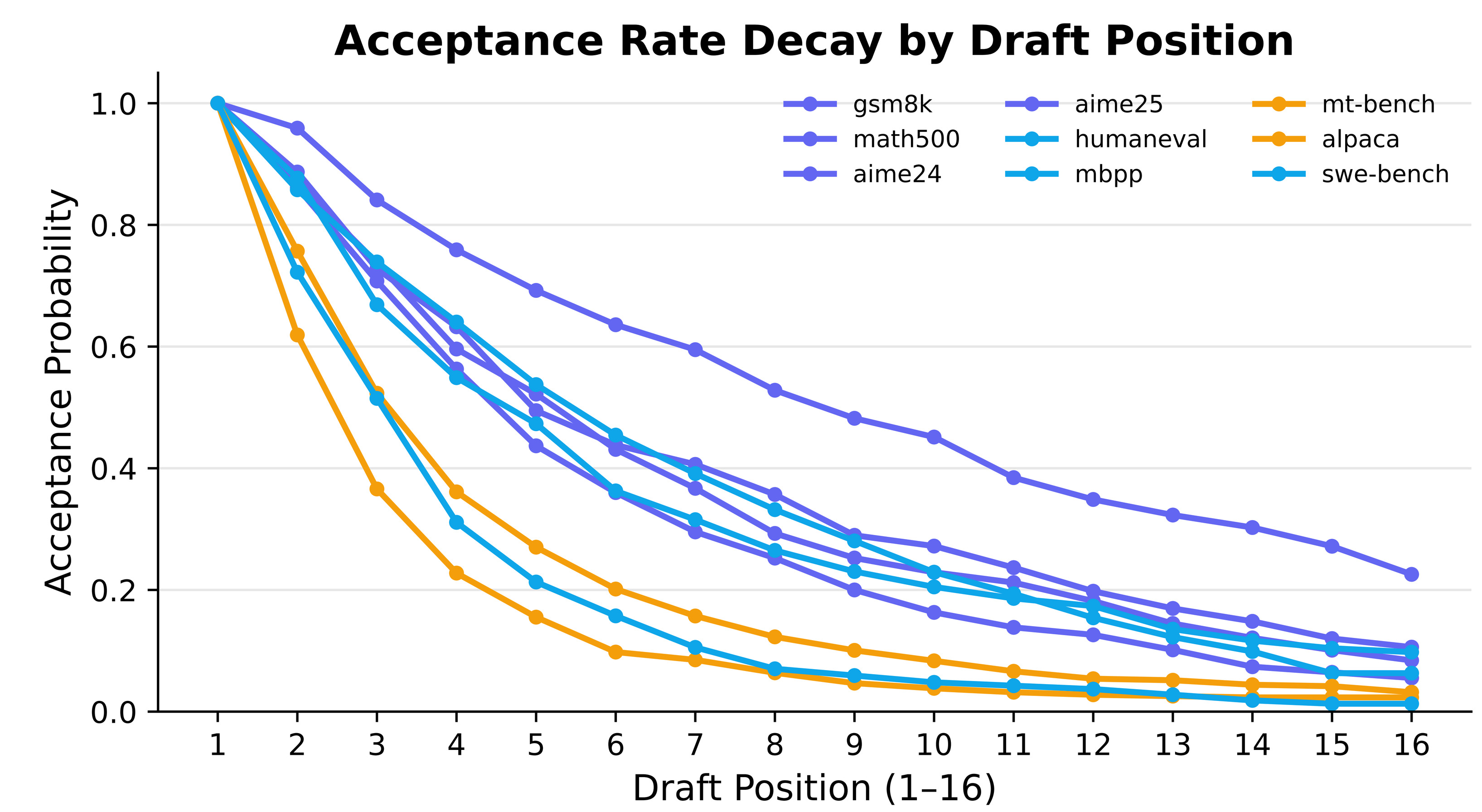


TPU vs GPU Draft Quality



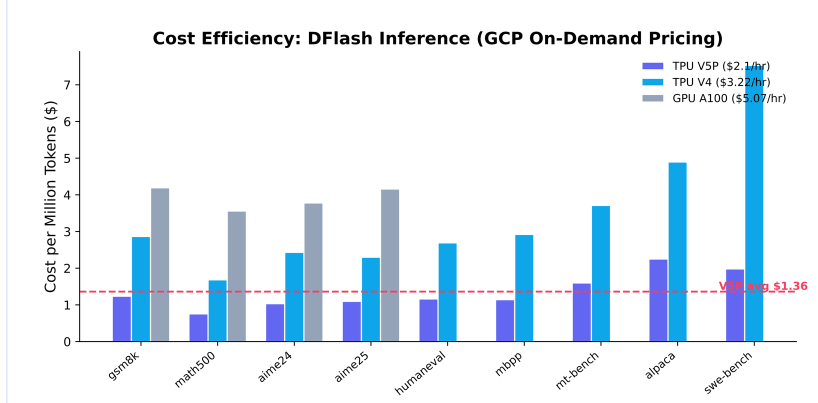
- **Draft quality parity:** TPU achieves 94.9% of GPU τ overall.
- Math500 *exceeds* GPU at 112%—the only benchmark where TPU surpasses the original GPU implementation.
- Remaining ∼5% gap attributed to bf16 precision and attention kernel differences.

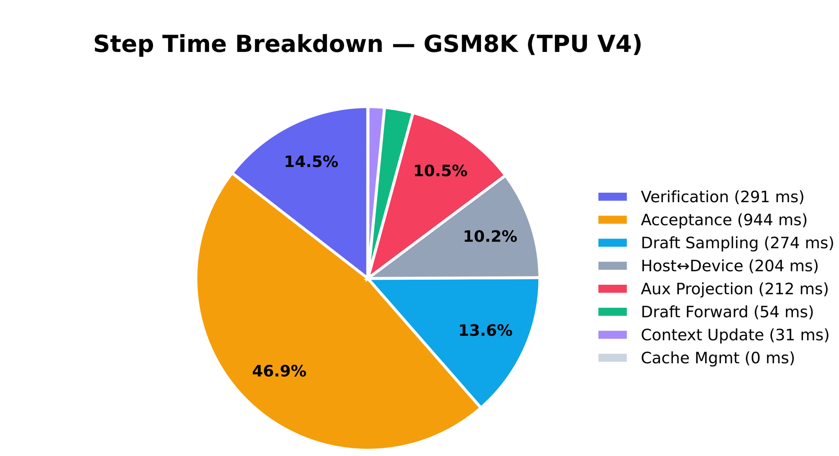
Acceptance Rate by Position



- Math benchmarks maintain >70% acceptance through position 8–10.
- Chat drops below 50% by position 4–5 due to higher entropy.
- Non-causal attention allows position 16 to "see" all 15 neighbors bidirectionally, preserving quality at later positions.

Conclusion & Future Work





- DFlash on TPU V5P: **3.01×** standalone, **2.31×** end-to-end pipeline speedup.
- Output quality is preserved—divergence is numerical (bf16), not errors.
- Pipeline overhead, not model compute, is the remaining bottleneck.
- TPU V5P + DFlash is the **most cost-efficient** option at $1.36/M tokens average.

Future Work

- **Wider blocks (K=64, 128):** Both draft and verify stay flat on TPU—should scale linearly.
- **vLLM optimization:** Close the standalone–pipeline τ gap (5.42 vs 4.48).
- **LM head fusion:** Two matmuls account for ∼30% of step time.

References & Acknowledgements

References

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